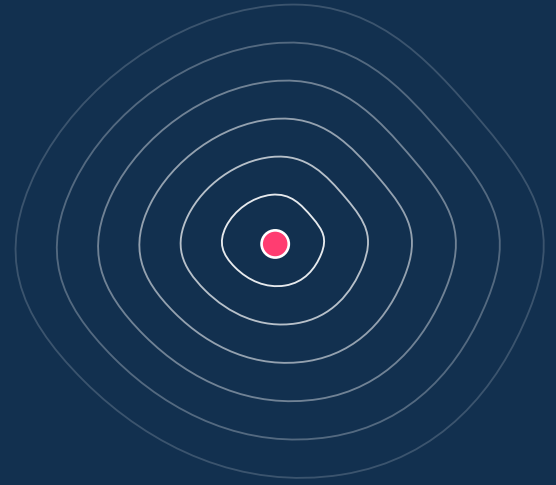


Village Pond Planning System

A geospatial decision-support web application for siting and sizing rainwater-harvesting ponds



Author **Amandeep Kammari**
Roll number **12341060**
Date **September 2026**
Source code <https://github.com/amandeeip01/village-pond-planner>
Live system <https://<your-app>.onrender.com>

In one sentence. Given only a village name, the system reconstructs the terrain, maps land that is free for excavation, delineates catchments, retrieves twenty years of satellite-gauge rainfall, simulates daily runoff and returns a ranked shortlist of pond sites with engineering dimensions — in about 40 seconds on first run and 3 seconds thereafter, using only free public data.

Abstract

Village ponds are among the most cost-effective rainwater-harvesting structures in rural India, but choosing *where* to dig one — on land that is actually available, on a drainage line that actually receives water, at a size the catchment can actually fill — normally requires a field survey and a hydrologist. This report presents a complete web application that automates this preliminary planning from open data. A digital elevation model is assembled from global terrain tiles (or interpolated from an uploaded KML contour map); surface flow is routed with Priority-Flood depression filling and the D8 algorithm; land availability is inferred by an OpenCV classifier on high-resolution satellite imagery combined with OpenStreetMap exclusions and optional government land parcels; candidate sites are scored on flow accumulation, slope, concavity and land suitability and then ranked so that each has an independent catchment; twenty years of daily CHIRPS rainfall drive an antecedent-moisture-adjusted SCS Curve Number runoff model; and a rectangular excavated pond is dimensioned to half of the 75 %-dependable annual runoff, subject to land and depth constraints. Results are delivered through a FastAPI back end and an accessible Leaflet front end with map overlays, charts, history and exports.

The system was validated component by component. Twenty-three unit tests check every formula against hand calculations and an analytic catchment reproduced cell for cell. Against IMD 1991–2020 station normals, the chosen rainfall source (CHIRPS) has a mean absolute error of 8.9 %, compared with 26.1 % for NASA POWER, which overestimated Pune by 92 %; this finding changed the system’s default data source. The land-cover classifier reached **92 % overall accuracy** ($\kappa = 0.85$) on 100 blind-labelled cells from four contrasting landscapes. A sensitivity analysis shows runoff has an elasticity of about three with respect to rainfall, identifying rainfall as the dominant uncertainty in pond sizing. For Hiware Bazar (Maharashtra), the system recommends five ponds with a combined storage of 121,155 m³, the best being a 141 m×94 m×3 m pond storing 36,856 m³ from a 138 ha catchment.

Keywords: rainwater harvesting; GIS; DEM hydrology; D8; SCS Curve Number; CHIRPS; OpenCV; land-cover classification; FastAPI; Leaflet.

Contents

Abstract	i
1 Introduction	1
1.1 Problem	1
1.2 Objective and scope	1
1.3 Contributions	1
2 Requirements	1
2.1 Functional requirements and traceability	1
2.2 Non-functional requirements	2
3 System design	2
3.1 High-level architecture	2
3.2 Analysis data flow	3
3.3 Request sequence	3
3.4 Data model	4
3.5 Key design decisions	4
4 Methodology	5
4.1 Terrain model	5
4.2 Flow routing	6
4.3 Land availability	6
4.4 Site scoring and ranking	7
4.5 Rainfall	8
4.6 Runoff	8
4.7 Pond design	9
5 Implementation	9
5.1 Code organisation	9
5.2 Performance	10
5.3 Robustness	10
5.4 Front end	11
6 Results: Hiware Bazar case study	12
6.1 Terrain and land	12
6.2 Rainfall	12
6.3 Candidate sites	13
6.4 Recommended pond (rank 1)	14
7 Verification and validation	14
7.1 Unit tests	14
7.2 Rainfall against IMD normals	15
7.3 Elevation	16
7.4 Land-cover classifier	17
7.5 Sensitivity analysis	17
7.6 What has not been validated	18
8 Software quality and project management	18

9	Limitations and future work	19
10	Conclusion	19
A	Installation guide	20
B	API reference	21

List of Figures

1	High-level architecture. Arrows show the direction of calls. Only the server talks to external services; all responses are cached.	3
2	Processing stages of one analysis. Rainfall is fetched once (at the top site) and shared by all candidates, since they lie within the same few kilometres.	3
3	Sequence of one village analysis.	4
4	SQLite schema. An index on <code>analyses(created_at)</code> serves the history list. . . .	4
5	Cross-section of the excavated pond, showing the design variables.	9
6	Results for Hiware Bazar: candidate 1 with its catchment (blue), drainage network, contours and candidates 2–5 (numbered markers). The sidebar shows the headline storage, indicators and the ranked candidate table.	11
7	Land-cover classification overlay (green: vegetation, yellow: open land, red: built-up) with candidate 1's catchment. The Hiware Bazaar settlement (lower right) is detected as built-up and excluded.	12
8	Hiware Bazar: (a) annual rainfall and simulated runoff depth, 2006–2025; (b) monthly climatology.	13
9	Candidate sites and the rainfall interface.	13
10	Rainfall datasets compared with IMD 1991–2020 station normals.	15
11	DEM compared with surveyed summit elevations: (a) scatter; (b) error distribution. .	16
12	Land-cover confusion matrix and per-site accuracy (87 unmixed cells).	17
13	Sensitivity of mean annual runoff at Hiware Bazar (rank 1): (a) to a uniform scaling of daily rainfall; (b) to hydrologic soil group.	18

List of Tables

1	Requirement traceability matrix.	2
2	Design decisions and rationale.	5
3	Curve numbers (AMC II) by land-cover class and hydrologic soil group, after TR-55. .	8
4	Back-end modules.	10
5	Stage timings, Hiware Bazar (cached run; first run adds ≈ 35 s of rainfall download). .	10
6	Ranked candidate ponds for Hiware Bazar (soil group C, CHIRPS 2006–2025). . .	14
7	Recommended pond, rank 1.	14
8	Unit tests and their independent references.	15
9	Mean annual rainfall 1991–2020 (mm) and error relative to the IMD normal.	16
10	HTTP API.	21

1 Introduction

1.1 Problem

Much of India's net sown area is rain-fed, and most of its rainfall arrives in the four months of the south-west monsoon. Small surface storages — farm ponds, village tanks, *Amrit Sarovars* — capture this water for irrigation, livestock and groundwater recharge. National programmes such as MGNREGA and Mission Amrit Sarovar fund thousands of such ponds a year [1], but the quality of each investment depends on three decisions taken early and often informally:

1. **Where?** The pond must sit on a drainage line so that runoff reaches it, on gentle ground so that excavation is stable, and on land the village is actually allowed to dig — normally gram-panchayat or government land.
2. **How much water will arrive?** This depends on the catchment area, its land cover and soils, and the rainfall regime — including dry years.
3. **How big?** A pond much larger than its dependable inflow rarely fills; one much smaller wastes the catchment. Depth trades off evaporation losses against excavation cost and safety.

1.2 Objective and scope

The objective, as set by the assignment, is a complete web application that analyses terrain and rainfall to recommend suitable pond locations, estimates catchment area, rainfall accumulation, pond depth and storage capacity, and presents the results on an interactive map. The system is a *pre-feasibility* tool: it narrows a village of several square kilometres down to a handful of defensible candidate sites, each with quantified inflow and dimensions, which an engineer can then verify in the field.

1.3 Contributions

- An end-to-end pipeline that runs from a place name to ranked, dimensioned pond sites using only free public data and no GIS server software (Sections 3–4).
- A land-availability model fusing OpenCV image classification, OpenStreetMap features and user-drawn government parcels (Section 4.3).
- A candidate-ranking algorithm that guarantees hydrologically independent alternatives (Algorithm 1).
- A daily, antecedent-moisture-aware SCS-CN runoff model and a constraint-driven pond design procedure (Sections 4.6–4.7).
- A quantitative validation of every data source and algorithm, which led to replacing the initial rainfall dataset (Section 7).

2 Requirements

2.1 Functional requirements and traceability

Table 1 maps each requirement in the brief to the modules, API fields and user interface elements that satisfy it.

Table 1: Requirement traceability matrix.

#	Requirement	Implementation	Evidence in output
1	Satellite imagery for a selected village	Nominatim geocoding (<code>geocode.py</code>); Esri World Imagery basemap with place labels	Village search box; satellite map
2	Visualise contour maps	OpenCV contour extraction from the DEM, or uploaded KML contours (<code>layers.py</code>); hillshaded relief	<code>layers.contours</code> , relief overlay
3	Identify available land	OpenCV classification + OSM exclusions + parcels + slope (<code>landcover.py</code>)	<code>land</code> , <code>layers.suitable_land</code>
4	Catchment area	Priority-Flood, D8, reverse traversal, boundary tracing (<code>hydrology.py</code>)	<code>catchment</code> , boundary GeoJSON
5	Historical rainfall via public APIs	CHIRPS (ClimateSERV), Open-Meteo, NASA POWER (<code>rainfall.py</code>)	<code>rainfall</code> : annual and monthly statistics
6	Runoff volume	Daily SCS-CN with AMC adjustment; rational check (<code>runoff.py</code>)	<code>runoff</code>
7	Pond depth and storage	Frustum design under land/depth constraints (<code>design.py</code>)	<code>pond_design</code> , cross-section
8	Overlay all results	Leaflet layers, charts, candidate table, exports (<code>static/</code>)	Map, sidebar, JSON/GeoJSON

2.2 Non-functional requirements

Generality

No coordinate, village or result may be hard-coded; every output must derive from the input location or file.

Cost

Only free, key-less public data services may be required.

Performance

A full analysis should complete within about a minute on first run and in seconds when cached.

Robustness

The failure of any single external service must degrade the result, not abort it, and the degradation must be reported to the user.

Transparency

Every number must be traceable to its data source and method; the API returns the methods, parameters and timings used.

Accessibility

The front end must be usable by keyboard and screen reader and meet WCAG AA contrast.

3 System design

3.1 High-level architecture

The system is a three-tier web application (Figure 1): a static single-page front end, a stateless FastAPI application server containing all analysis logic, and a persistence tier consisting of an

SQLite database and an on-disk tile cache. External data services are accessed only from the server, which lets it set proper User-Agent headers, cache responses and keep API usage within fair-use limits.

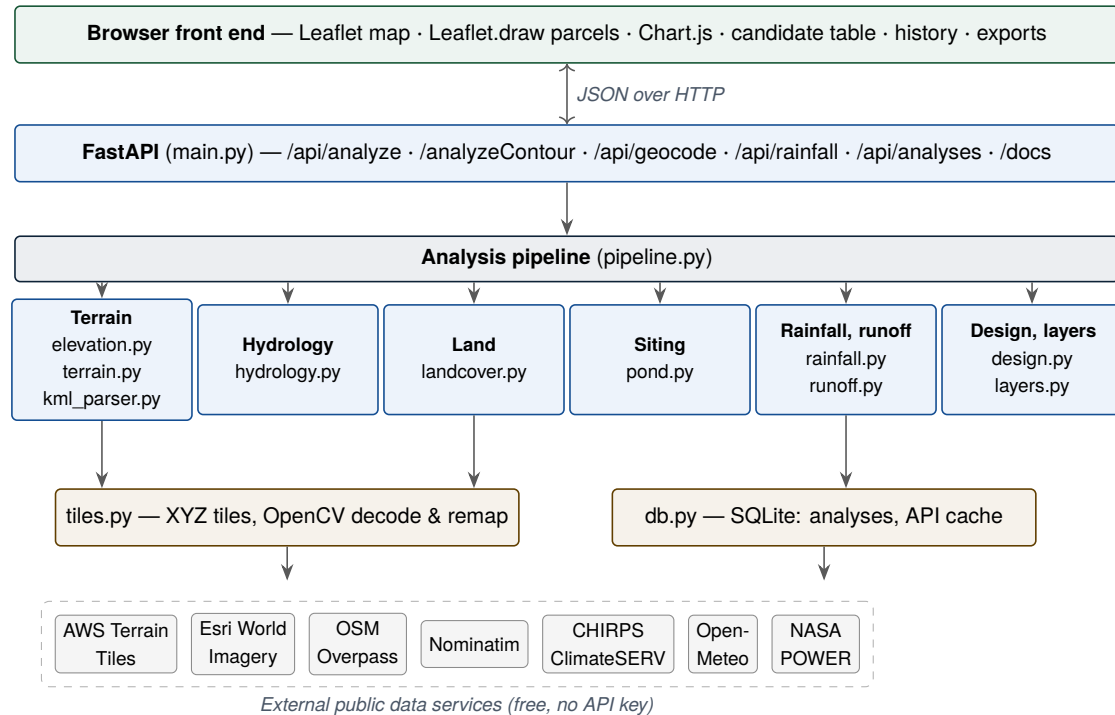


Figure 1: High-level architecture. Arrows show the direction of calls. Only the server talks to external services; all responses are cached.

3.2 Analysis data flow

Figure 2 shows the processing stages of one analysis. The two input paths — village location or uploaded contour map — converge on a common in-memory DEM, after which the pipeline is identical. This separation is deliberate: every downstream algorithm depends only on a regular elevation grid and a projection, so new terrain sources (GeoTIFF, drone surveys) can be added without touching the hydrology.

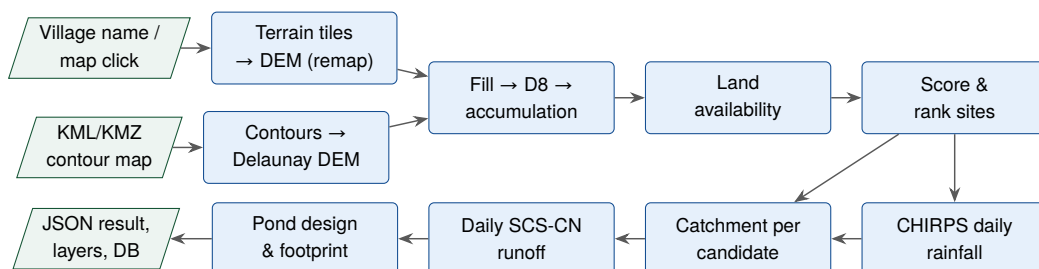


Figure 2: Processing stages of one analysis. Rainfall is fetched once (at the top site) and shared by all candidates, since they lie within the same few kilometres.

3.3 Request sequence

A village analysis is a single synchronous `POST /api/analyze` request. Within it, the slow rainfall download (CHIRPS, two ten-year chunks in parallel) is overlapped with the evapotranspiration

download from a second service (Figure 3). On a repeat request for the same area every external call is served from the tile cache or the `api_cache` table.

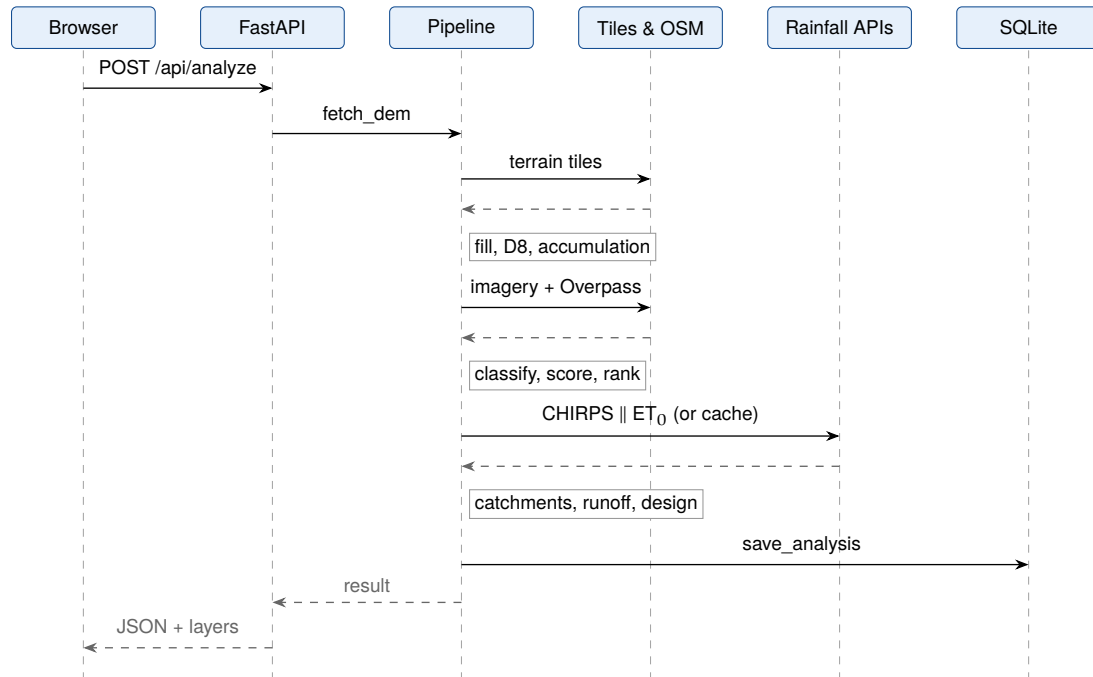


Figure 3: Sequence of one village analysis.

3.4 Data model

Persistence is intentionally light (Figure 4). The `analyses` table stores each completed analysis twice: a compact summary for the history list and the full result document for exact re-display. The `api_cache` table memoises external responses (rainfall series, geocoding) with a time-to-live. Tiles are cached as files keyed by `source/z/x/y`.

analyses	api_cache
id INTEGER PK autoincrement	key TEXT PK
created_at TEXT (UTC ISO-8601)	e.g.
village TEXT	rain:chirps:19.08:74.60:2006:2025
mode TEXT (village contour_upload)	created_at REAL (epoch s)
lat, lon REAL (top pond site)	payload TEXT (JSON)
summary_json TEXT (history card)	TTL: rainfall 30 days, geocoding 7 days
result_json TEXT (full response)	

Figure 4: SQLite schema. An index on `analyses(created_at)` serves the history list.

3.5 Key design decisions

Table 2 records the main design choices, the alternatives considered and the reasons. They are the decisions most likely to be questioned, so each is justified by a requirement or by measured evidence.

Table 2: Design decisions and rationale.

Decision	Alternatives	Rationale
FastAPI	Flask, Django	Pydantic models give request validation and an OpenAPI/Swagger reference for free (deliverable: API documentation); async-capable; the same schemas document and validate.
NumPy / SciPy / OpenCV hydrology, no GDAL	GDAL, WhiteboxTools, GRASS	Installs from pip wheels on any platform and on free hosting; every algorithm is visible and testable. The algorithms are standard and small enough to implement directly.
Terrain tiles for the DEM	OpenTopography, Open-Elevation, SRTM downloads	One HTTP request per 256×256 tile, global, key-less and cacheable. A point-sampling API would need 40,000 queries for a 200×200 grid.
Local equirectangular projection	UTM via pyproj	Distortion is far below the DEM error over a few kilometres; avoids a PROJ dependency.
CHIRPS as default rainfall	Open-Meteo ERA5, NASA POWER, IMD grids	Best agreement with IMD normals (Section 7.2); IMD grids are bulk files, not an API. The others are kept as fallbacks.
Daily SCS-CN	Rational method, monthly water balance	Captures storm-size non-linearity and antecedent moisture; the standard in Indian watershed manuals. Rational method kept as a cross-check.
OpenCV rules + OSM for land	Deep segmentation model	No labelled training data for Indian villages; rule-based indices are explainable, need no GPU, and reached 92 % accuracy (Section 7.4).
SQLite	PostgreSQL, MongoDB	The data is a small document store with no concurrent writers; zero administration. The <code>db.py</code> interface isolates the choice.
Leaflet + vanilla JS	React, Mapbox GL	The brief asks for a basic front-end library; no build step; small bundle.

4 Methodology

4.1 Terrain model

Grid and projection. For a village centre (λ_0, φ_0) and radius R (default 2 km) a square grid of $N \times N$ cells (default $N = 200$, cell size $\Delta = 2R/N = 20$ m) is defined in a local equirectangular projection:

$$x = R_E \cos \varphi_0 (\lambda - \lambda_0) \frac{\pi}{180}, \quad y = R_E (\varphi - \varphi_0) \frac{\pi}{180}, \quad (1)$$

with $R_E = 6,371,008.8$ m. Row 0 is the southern edge.

Elevation. AWS Terrain Tiles [2] encode elevation in the RGB bytes of Web-Mercator PNG tiles:

$$z = 256R + G + B/256 - 32768 \quad [\text{m}]. \quad (2)$$

The tiles covering the grid are downloaded in parallel, decoded with OpenCV and stitched. The elevation mosaic is then sampled at every cell centre with `cv2.remap` using bilinear interpolation. Decoding happens *before* resampling: interpolating the packed bytes would mix the high and low bytes of neighbouring pixels and produce errors of hundreds of metres.

Contour-map input. For uploaded KML/KMZ maps, the elevation of each contour is detected by trying four strategies in order: coordinate z values, `ExtendedData`, the placemark name, then

the folder name. A strategy is accepted only if it covers at least 90 % of lines and yields varying values. The contour vertices are then linearly interpolated over their Delaunay triangulation, with nearest-neighbour fill outside the convex hull.

4.2 Flow routing

Depression filling. Interpolated DEMs contain spurious pits that trap routed water. The Priority-Flood algorithm [3] seeds a priority queue with all edge cells and repeatedly pops the lowest cell, raising each unvisited neighbour to at least the popped elevation plus $\varepsilon = 10^{-4}$ m. This yields a surface in which every cell has a monotone downhill path to the edge, in $O(n \log n)$.

D8 directions. Each cell drains to the neighbour k maximising the distance-weighted drop [4]

$$s_k = \frac{z_c - z_k}{d_k}, \quad d_k \in \{1, \sqrt{2}\} \Delta, \quad (3)$$

and is marked as draining off-grid if no $s_k > 0$. The computation is vectorised as eight shifted array comparisons.

Flow accumulation. Cells are visited in decreasing filled elevation; each adds its count to its downstream neighbour. Because flow is strictly downhill on the filled surface, a cell's total is final before it is passed on. This removes any need for recursion or graph sorting.

Catchment delineation and boundary. The catchment of an outlet is found by breadth-first traversal of the *reversed* D8 graph. Its boundary is traced by collecting every cell edge that separates catchment from non-catchment, stitching the edges into a ring, and dropping collinear vertices. The polygon therefore follows exact cell edges, and its shoelace area equals the cell count times Δ^2 , which a unit test verifies.

Edge-inflow detection. A catchment that extends beyond the map would be underestimated. A border cell whose D8 direction points *into* the grid sits on ground sloping inward from the edge, so land beyond the edge probably drains in. All cells downstream of such a cell are flagged by a highest-first propagation analogous to accumulation. In village mode flagged cells are excluded from siting, which guarantees complete catchments.

4.3 Land availability

Three independent evidence layers are rasterised onto the analysis grid.

Satellite classification (OpenCV). Esri World Imagery is resampled to a grid four times finer than the DEM (5 m). Each pixel is classified as follows.

- *Vegetation:* pixels where the Green–Red Vegetation Index $GRVI = (G - R)/(G + R)$ [5, 6] exceeds a threshold τ . The threshold is chosen per image by Otsu's method [7] and clamped to $[0, 0.10]$ so that an almost uniformly bare or green scene is not split in half.
- *Built-up:* clusters of bright, highly textured pixels. A pixel is a candidate if its local standard deviation over 15 m exceeds $\max(18, P_{90})$ and its brightness exceeds the median. It is built-up if the candidate density over a 40 m window exceeds 0.30. The density test separates villages from isolated field boundaries.
- *Water:* dark ($V < 90$), blue-dominant and smooth pixels.

- *Open land*: everything else.

Morphological opening and closing (5 px ellipse) remove speckle. Class fractions per DEM cell are obtained by area-averaging with `INTER_AREA`.

OpenStreetMap. One Overpass query retrieves buildings, roads, railways, waterways, land use, natural features, amenities and protected areas. Features are burned into masks with `cv2.fillPoly` and `cv2.polylines`, with safety buffers: 20 m around buildings and 4 m to 30 m around roads depending on class. Multipolygon relations are stitched from their member ways. Settlements, institutions, forest, water bodies and rivers are excluded. Grassland, scrub, meadow, village green or public-ownership tags mark land as *preferred*, since in Indian villages such land is usually common or government land. Farmland remains allowed but is ranked lower.

Parcels and slope. If the administrator draws or uploads government land parcels, siting is restricted to them. Finally, ground steeper than a threshold (default 8°) is excluded. The result is a boolean *allowed* mask, a land tier (preferred / open / farmland / excluded) and a land score in $[0, 1]$.

4.4 Site scoring and ranking

Within the interior of the grid (excluding a 6 % border and edge-inflow cells), the candidates are allowed cells on the drainage network. The drainage threshold is the 92nd percentile of accumulation, relaxed in steps to the 60th if available land does not reach the main channels. Each candidate c receives the score

$$S_c = 0.75(0.45 \hat{A}_c + 0.35 (1 - \hat{\theta}_c) + 0.20 \hat{\kappa}_c) + 0.25 L_c, \quad (4)$$

where $\hat{A}$ is the min-max normalised $\log(1 + \text{accumulation})$, $\hat{\theta}$ the normalised slope, $\hat{\kappa}$ the normalised concavity (negative Laplacian) and L the land score. The weights encode the priorities of pond siting: water must arrive, excavation must be stable, and hollows hold water naturally.

Taking the top five scores would return five adjacent cells on one stream. Algorithm 1 instead enforces spatial separation and hydrological independence: no candidate may lie upstream or downstream of another, so each pond harvests a different part of the landscape and their inflows can be summed without double counting.

Algorithm 1 Ranking independent candidate sites

Require: score grid S , D8 directions D , number n , minimum separation $d_{\min}$

```

1:  $A \leftarrow []$ ;  $M \leftarrow []$  ▷ accepted sites, their catchment masks
2: for all cells  $c$  in decreasing order of  $S$  do
3:   if  $|A| = n$  then break
4:   if  $\exists a \in A : \|c - a\| < d_{\min}$  then continue ▷ too close
5:   if  $\exists m \in M : m[c]$  then continue ▷  $c$  upstream of an accepted site
6:    $m_c \leftarrow \text{CATCHMENT}(D, c)$ 
7:   if  $\exists a \in A : m_c[a]$  then continue ▷  $c$  downstream of an accepted site
8:   append  $c$  to  $A$  and  $m_c$  to  $M$ 
9: end for
10: return  $A$ 

```

A user-chosen location is snapped to the highest-accumulation cell within a 60 m radius (on allowed land if possible). It becomes rank 1, and the automatic sites follow as alternatives.

4.5 Rainfall

Sources. Precipitation is taken, in order of preference, from:

1. CHIRPS v2.0 [8], a 0.05° daily product that blends satellite estimates with rain-gauge data, accessed through NASA/USAID SERVIR ClimateSERV in parallel ten-year chunks;
2. Open-Meteo's ERA5 reanalysis [9];
3. NASA POWER [10].

Reference evapotranspiration ET_0 comes from Open-Meteo (FAO-56 Penman–Monteith) or is computed from NASA POWER temperatures by Hargreaves–Samani [11]:

$$ET_0 = 0.0023 (0.408R_a) (T_{\text{mean}} + 17.8) \sqrt{T_{\text{max}} - T_{\text{min}}}, \quad (5)$$

with extraterrestrial radiation R_a from FAO-56 eq. 21 [12]. If the administrator enters a known long-term normal $\bar{P}^*$, the daily series is linearly scaled by $\bar{P}^*/\bar{P}$. This preserves the storm pattern while removing the bias.

Statistics. The following statistics are computed from the daily series: mean, median, standard deviation and coefficient of variation of annual totals; the 75 %-dependable rainfall (25th percentile of annual totals, i.e. equalled or exceeded three years in four); the driest and wettest years; rainy days (≥ 2.5 mm, the IMD definition); the heaviest day; the monsoon share (June–September); monthly climatology; and annual and dry-season (October–May) ET_0 .

4.6 Runoff

Runoff is simulated for every day of the record with the SCS Curve Number method [13]:

$$S = \frac{25400}{CN} - 254, \quad I_a = 0.2S, \quad Q = \begin{cases} \frac{(P - I_a)^2}{P - I_a + S}, & P > I_a \\ 0, & \text{otherwise} \end{cases} \quad [\text{mm}]. \quad (6)$$

Curve numbers (Table 3) are assigned per land-cover class for the chosen hydrologic soil group. Runoff is computed per class and area-weighted over the catchment's land-cover fractions. This is more accurate than applying a single composite CN, because the equation is non-linear. Each day's CN is adjusted for antecedent moisture from the preceding five days' rainfall P_5 : AMC I if $P_5 < 35.6$ mm, AMC III if $P_5 > 53.3$ mm, using Hawkins' conversions [14]

$$CN_I = \frac{CN}{2.281 - 0.01281 CN}, \quad CN_{III} = \frac{CN}{0.427 + 0.00573 CN}. \quad (7)$$

Annual volumes are $V = Q \cdot A$. The mean and the 75 %-dependable (25th percentile) annual volumes are reported. The rational method $V = A\bar{P}C$, with a land-cover-weighted C , is reported alongside as a cross-check.

Table 3: Curve numbers (AMC II) by land-cover class and hydrologic soil group, after TR-55.

Class	TR-55 cover type	A	B	C	D	Rational C
Water	rain on water surface	98	98	98	98	1.00
Built-up	residential, $\approx 85\%$ impervious	89	92	94	95	0.75
Vegetation	row crops, straight row, good	67	78	85	89	0.25
Open land	pasture/fallow, poor	68	79	86	89	0.35
Forest (OSM)	woods, fair	36	60	73	79	0.15

4.7 Pond design

The pond is a rectangular excavated frustum with side slope $zH:1V$ (default 1.5), length $1.5\times$ width, with the long side laid along the contour, i.e. perpendicular to the local gradient. For top width W , length L and depth d , the volume is given exactly by the prismoidal formula

$$V = \frac{d}{6} (LW + 4(L - zd)(W - zd) + (L - 2zd)(W - 2zd)), \quad (8)$$

and W is found for a target V by bisection. The design procedure is:

1. **Target storage** $V^* = f \cdot V_{75}$, with $f = 0.5$ by default. The pond then fills about twice in a dependable (dry) year, leaving margin for evaporation, seepage and a weak early monsoon.
2. **Depth.** Start at 3 m, the depth used in MGNREGA and Amrit Sarovar designs [1]. If the footprint (including 0.5 m freeboard) exceeds 80 % of the connected patch of available land at the site, or the 2 ha village-pond cap, increase the depth in 0.5 m steps up to 4.5 m. Deeper ponds also lose proportionally less water to evaporation.
3. **Capping.** If the pond still does not fit, reduce the volume to what fits and say so.
4. **Small ponds.** Reduce the depth (down to 2 m) so that the bed stays at least 4 m wide for machinery.

The procedure also reports:

- excavation volume;
- dry-season open-water evaporation $E = 1.05 ET_{0, \text{Oct-May}} \cdot A_{\text{mid}}$, where A_{mid} is the water surface at half depth [12];
- four-month seepage at a rate that depends on soil group (1–10 mm/day), with lining advice for soil groups A and B;
- fills per dependable year.

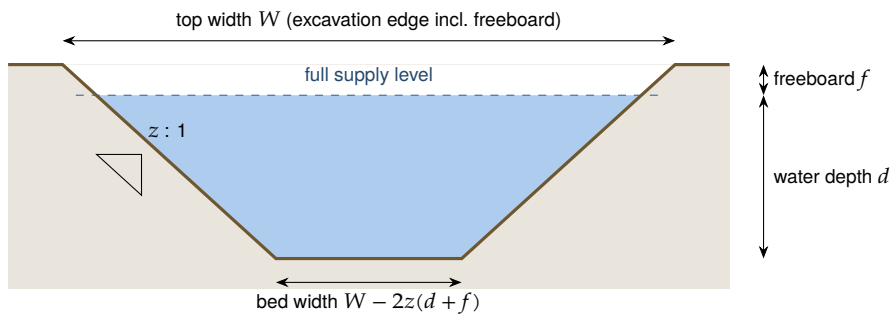


Figure 5: Cross-section of the excavated pond, showing the design variables.

5 Implementation

5.1 Code organisation

The back end comprises about 3,400 lines of Python in fourteen modules (Table 4). The front end is about 1,200 lines of HTML, CSS and JavaScript, and there are about 220 lines of tests. Modules are layered: `main.py` handles only HTTP concerns; `pipeline.py` orchestrates; each analysis module is a pure function of arrays and parameters and is independently testable; I/O is confined to `tiles.py`, `rainfall.py`, `landcover.fetch_osm`, `geocode.py` and `db.py`.

Table 4: Back-end modules.

Module	Lines	Responsibility
main.py	236	FastAPI routes, request validation, error mapping, static front end
schemas.py	98	Pydantic request/response models; drive the OpenAPI documentation
pipeline.py	334	Orchestration; per-candidate evaluation; result assembly; persistence
tiles.py	164	Web-Mercator maths, parallel tile download, disk cache, OpenCV mosaic and remap
elevation.py	91	DEM from terrain tiles, with Open-Meteo elevation fallback
terrain.py	197	Projection, DEM grid, contour interpolation, slope and curvature
kml_parser.py	225	KML/KMZ parsing with four elevation-detection strategies
hydrology.py	239	Priority-Flood, D8, accumulation, edge inflow, catchment, boundary
landcover.py	457	OpenCV classification, Overpass query, rasterisation, parcels, tiers
pond.py	259	Site scoring, candidate ranking, snapping, natural-basin storage
rainfall.py	340	CHIRPS/Open-Meteo/NASA clients, parallel fetch, cache, scaling, statistics
runoff.py	150	Curve numbers, AMC adjustment, daily SCS-CN, rational method
design.py	184	Frustum geometry, depth selection, losses, footprint polygon
layers.py	262	Contours, drainage network, land polygons, PNG overlays
db.py	112	SQLite persistence and API cache

5.2 Performance

Table 5 shows the stage timings for Hiware Bazar on a laptop. Terrain analysis on 40,000 cells takes about 60 ms because D8 is vectorised and accumulation is a single sorted pass. The first run for a location is dominated by the CHIRPS download (about 35 s for two parallel ten-year requests). Repeat runs use the cache and take about 2.5 s.

Table 5: Stage timings, Hiware Bazar (cached run; first run adds ≈ 35 s of rainfall download).

Stage	Time (s)
Fill, D8, accumulation, edge inflow	0.062
Land assessment (imagery classification + OSM)	1.524
Scoring and ranking 5 candidates	0.066
Rainfall (cache)	0.752
Catchments, runoff and design for 5 candidates	0.047
Map layers	0.022
Total server time	2.47

5.3 Robustness

Every external dependency has a defined failure mode.

- Terrain tiles fall back to the Open-Meteo elevation API.
- Rainfall falls back through three providers. Transient network errors are retried with back-off; definitive errors such as rate limits move straight to the next source.
- If imagery or OpenStreetMap is unavailable, siting continues on the remaining evidence.
- If all rainfall services fail, the rational method is applied to a fallback rainfall value.

In every case the degradation appears in the response's `warnings` array and on screen. Validation errors return HTTP 422, upstream outages 502, and all errors share the JSON shape `{"status": "error", "detail": ...}`.

5.4 Front end

The front end is a single page with Village, Contour file and History tabs. The Village tab provides:

- village search;
- a preview of the analysis area;
- parcel drawing and upload (Leaflet.draw);
- an optional manual pond pin;
- advanced parameters: soil group, rainfall dataset and normal, storage fraction and depth limits.

While an analysis runs, a staged progress overlay covers the map. The results comprise:

- a headline storage card and four key indicators;
- the ranked candidate table, also shown as numbered map markers, where selecting a candidate re-renders everything for it;
- Pond, Rainfall, Runoff and Site & land sub-tabs with a pond cross-section and charts;
- map overlays for contours, drainage, catchment, available land, land-cover and relief rasters, and pond footprints;
- JSON and GeoJSON export and a print stylesheet for a paper report.

Accessibility work included ARIA tab semantics, keyboard-operable lists and table rows, visible focus outlines, a skip link, labelled map and chart regions, a reduced-motion mode, and a text colour raised to a 5.6:1 contrast ratio (WCAG AA).

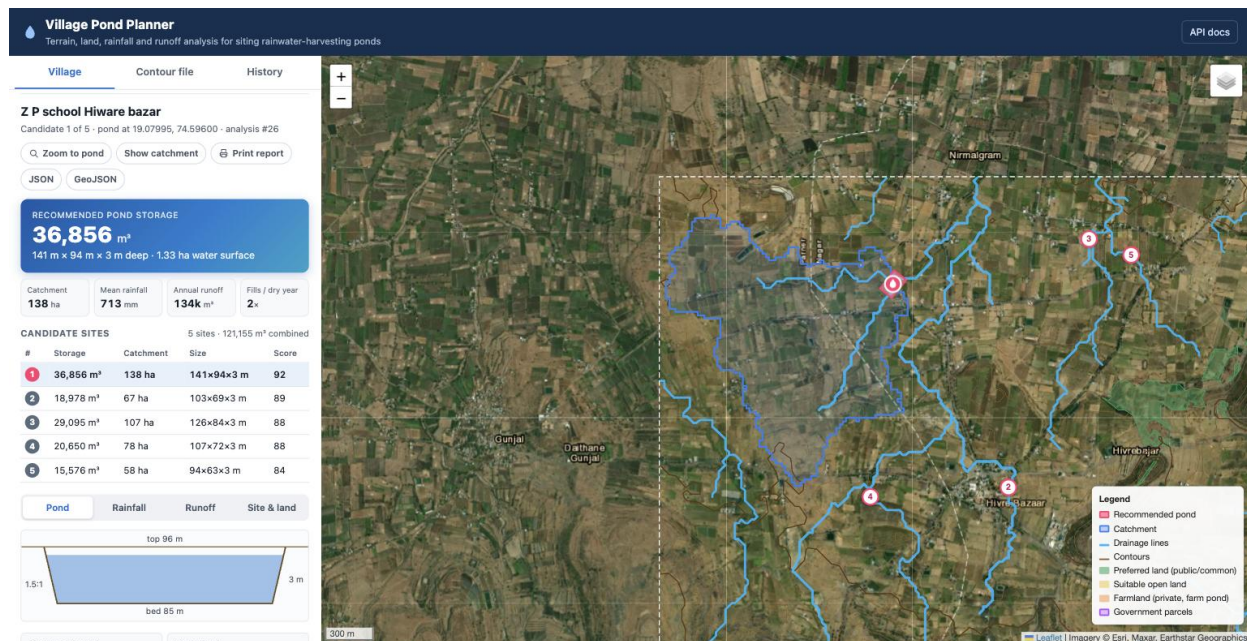


Figure 6: Results for Hiware Bazar: candidate 1 with its catchment (blue), drainage network, contours and candidates 2–5 (numbered markers). The sidebar shows the headline storage, indicators and the ranked candidate table.

6 Results: Hiware Bazar case study

Hiware Bazar (Ahilyanagar district, Maharashtra) is a well-known model village for watershed development on the semi-arid Deccan plateau, in the rain shadow of the Western Ghats. The analysis covered a 4 km×4 km area centred at 19.0680° N, 74.6012° E, with soil group C and default parameters.

6.1 Terrain and land

The DEM spans 692 m to 875 m. OpenStreetMap supplied 111 buildings, 39 roads and 23 water features. After exclusions and the 8° slope limit, 76 % of the area (1,219 ha) is suitable, of which 24.4 ha carries tags that suggest common land. Figure 7 shows the classification: the village core is correctly identified as built-up, and irrigated plots as vegetation within largely open, fallow terrain.

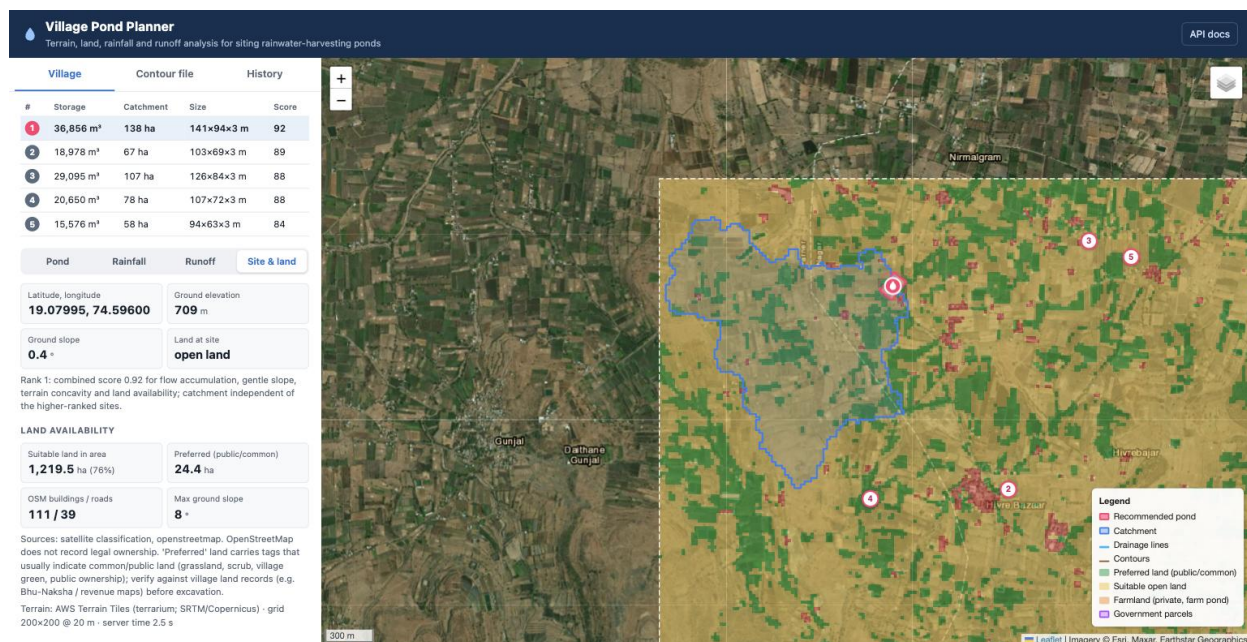


Figure 7: Land-cover classification overlay (green: vegetation, yellow: open land, red: built-up) with candidate 1's catchment. The Hiware Bazaar settlement (lower right) is detected as built-up and excluded.

6.2 Rainfall

CHIRPS gives a mean of 713 mm for 2006–2025 (Figure 8). The 75 %-dependable rainfall is 599 mm, and the coefficient of variation of 23 % shows typical Deccan variability: from 373 mm in 2018 to 1,063 mm in 2020. The site records 53 rainy days a year, and 79 % of the rain falls between June and September. The heaviest day recorded is 78 mm. Reference evapotranspiration is 1,771 mm/yr, of which 1,249 mm falls in the October–May dry season.

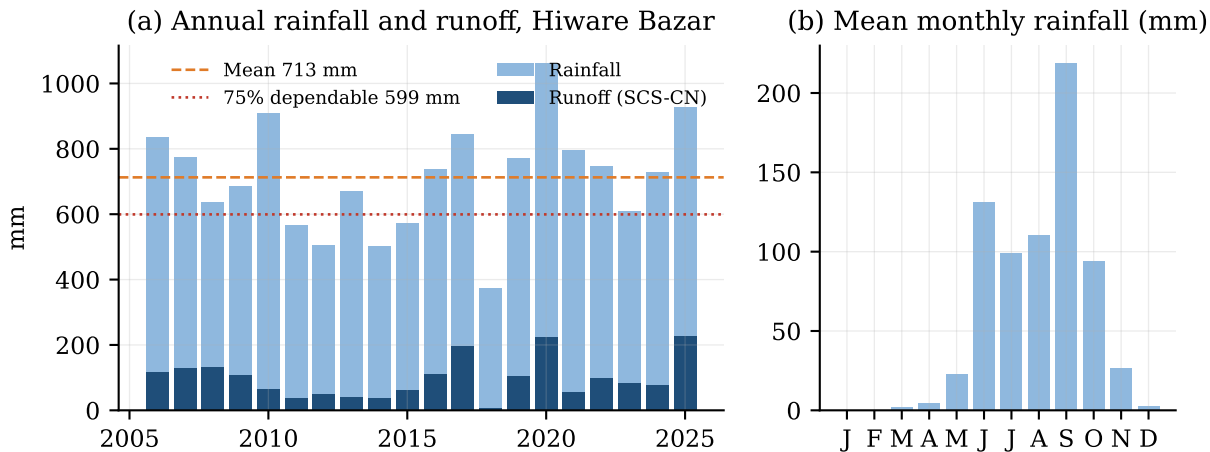
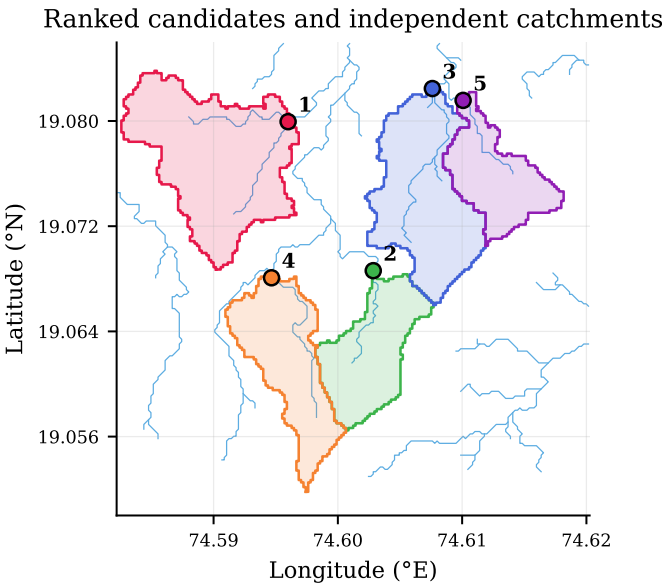


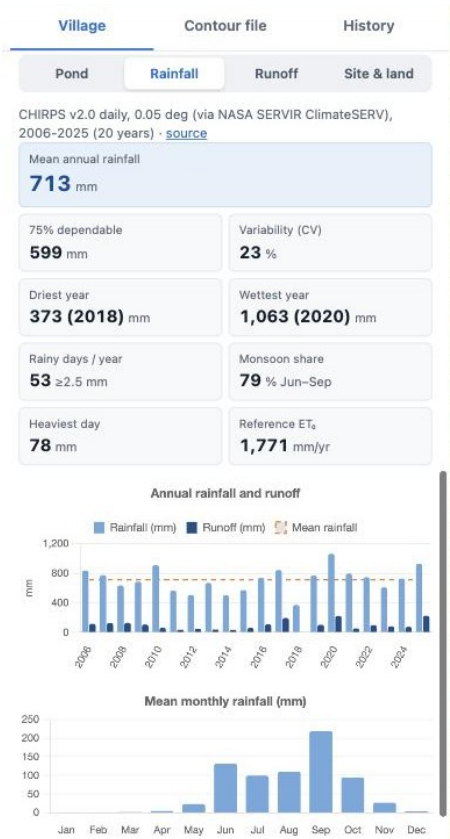
Figure 8: Hiware Bazar: (a) annual rainfall and simulated runoff depth, 2006–2025; (b) monthly climatology.

6.3 Candidate sites

Algorithm 1 found five hydrologically independent sites (Figure 9, Table 6), all on open land with slopes of 0.1°–1.6°. Their combined storage is 121,155 m³.



(a) Five ranked sites and their catchments.



(b) Rainfall tab of the user interface.

Figure 9: Candidate sites and the rainfall interface.

Table 6: Ranked candidate ponds for Hiware Bazar (soil group C, CHIRPS 2006–2025).

Rank	Score	Location (°N, °E)	Catch. (ha)	Mean runoff (m ³ /yr)	Dep. runoff (m ³ /yr)	Storage (m ³)	Top size (m)	Earthwork (m ³)
1	0.92	19.07995, 74.59600	137.6	134,368	73,712	36,856	141×94	43,604
2	0.89	19.06862, 74.60285	67.3	68,007	37,955	18,978	103×69	22,578
3	0.88	19.08247, 74.60760	106.5	105,393	58,190	29,095	126×84	34,483
4	0.88	19.06808, 74.59466	78.5	75,657	41,301	20,650	107×72	24,548
5	0.84	19.08157, 74.61008	57.7	56,694	31,153	15,576	94×63	18,570
Total			447.6	440,119		121,155		143,782

6.4 Recommended pond (rank 1)

The top site lies at 19.07995° N, 74.59600° E. The ground is at 709 m with a slope of 0.4°, on open land at the outlet of a 137.6 ha catchment (perimeter 7.72 km, relief 44.8 m, mean slope 3.3°). The catchment is 77 % open or fallow land, 22 % crops and 1 % built-up, giving a composite CN of 85.8. Mean runoff is 98 mm/yr, an effective coefficient of 0.14, or 134,368 m³/yr. In a dependable year it is 73,712 m³. Table 7 summarises the design.

Table 7: Recommended pond, rank 1.

Storage capacity	36,856 m³	Water spread area	1.33 ha
Top size (L×W)	141.3 m×94.2 m	Excavation volume	43,604 m ³
Bed size (L×W)	132.3 m×85.2 m	Dry-season evaporation	16,099 m ³ (44 %)
Water depth + freeboard	3.0 m + 0.5 m	Seepage (4 months)	3,384 m ³
Side slope	1.5:1	Fills per dependable year	2.0

Interpretation

Losing 44 % of storage to evaporation over the dry season is realistic for the Deccan, where October–May ET₀ exceeds 1.2 m. This is why the design favours depth over surface area once land becomes limiting. The five candidates together could capture about 28 % of the mean runoff generated by their catchments. This leaves most of the flow to continue downstream, which avoids starving downstream users.

7 Verification and validation

Verification (*is the model implemented correctly?*) was done with unit tests. Validation (*does the model match reality?*) was done by comparing each data source and classifier with an independent reference. All scripts, raw results and labelled imagery are in the `validation/` directory and can be re-run.

7.1 Unit tests

Twenty-three `pytest` tests pass in 0.2 s (Table 8). The most informative is the synthetic V-valley: two planes meet along a channel sloping to the south. Off-channel cells step diagonally, so a cell at row r and distance k from the channel joins it at row $r - k$. The analytic catchment of the outlet at row r_0 is therefore $\{(r, c) : r - |c - c_0| \geq r_0\}$. The implementation reproduces this cell for cell.

Table 8: Unit tests and their independent references.

Test	Reference
SCS runoff, $P = 100$ mm, CN 80	Hand calculation: $S = 63.5$, $I_a = 12.7$, $Q = 87.3^2 / 150.8 = 50.54$ mm
SCS, four further (P , CN) pairs; zero below I_a	Closed-form TR-55 equation
AMC conversion	Hawkins (1985): CN 75 $\rightarrow$ 56.8 (I), 87.0 (III)
Dry vs wet antecedent conditions	$Q_I < Q_{II} < Q_{III}$
Runoff volume; rational fallback	$Q \cdot A$; $A P C$
Frustum volume (square, rectangle)	$\frac{h}{3}(A_1 + A_2 + \sqrt{A_1 A_2}) = 650.67 \text{ m}^3$; prismoidal formula by hand
Width solver	Round trip at three volumes
Pond rules	Hits target; deeper on small land; caps storage and reports it
Extraterrestrial radiation	FAO-56 Example 8: 20°S , 3 Sep $\rightarrow 32.2 \text{ MJ m}^{-2} \text{ d}^{-1}$
Rainfall statistics	Constructed series with known mean, percentile, driest year, rainy days, ET_0
Depression filling	No interior sinks; surface never lowered
V-valley catchment	Analytic set, cell for cell; accumulation equals its size
Boundary polygon	Shoelace area = cell count
Bowl with channel; edge inflow	All cells reach the outlet; inward valley flagged, hill-top not

7.2 Rainfall against IMD normals

For six IMD observatories in contrasting climates, each dataset's 1991–2020 mean annual rainfall was compared with the IMD 1991–2020 normal (Figure 10, Table 9).

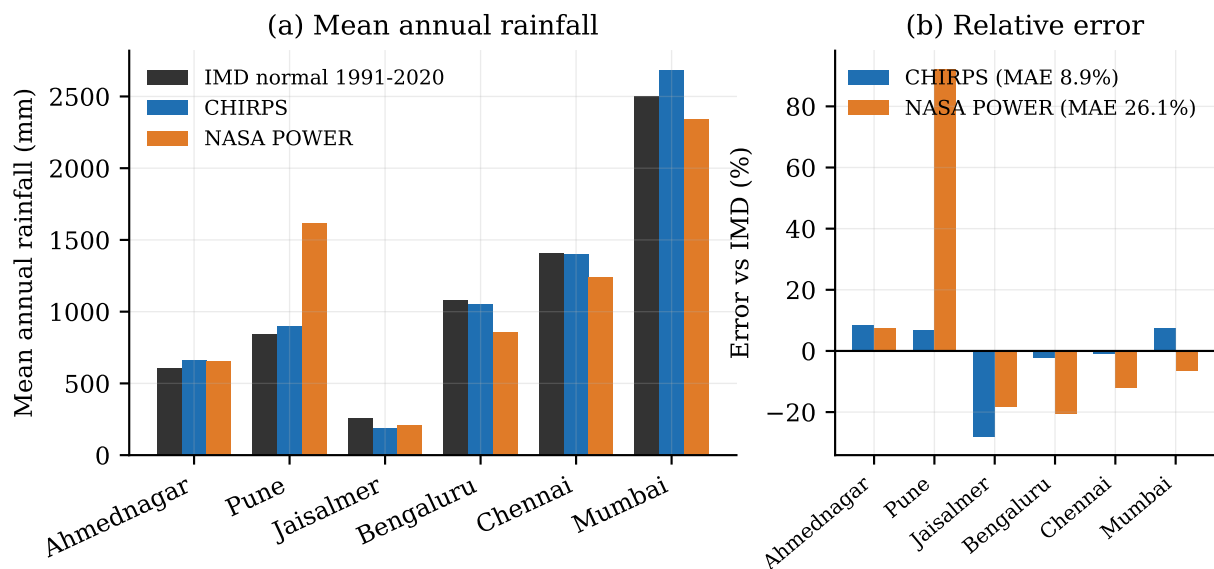
**Figure 10:** Rainfall datasets compared with IMD 1991–2020 station normals.

Table 9: Mean annual rainfall 1991–2020 (mm) and error relative to the IMD normal.

Station	Climate	IMD	CHIRPS	Error	NASA POWER	Error
Ahmednagar	semi-arid, rain shadow	608.1	658.1	+8.2 %	652.0	+7.2 %
Pune ^a	leeward Western Ghats	841.2	897.9	+6.7 %	1616.7	+92.2 %
Jaisalmer	arid	256.9	184.8	−28.1 %	210.1	−18.2 %
Bengaluru	semi-arid plateau	1077.8	1054.0	−2.2 %	858.7	−20.3 %
Chennai	NE-monsoon coast	1408.4	1396.8	−0.8 %	1237.9	−12.1 %
Mumbai	windward coast	2502.3	2684.6	+7.3 %	2341.7	−6.4 %
Mean absolute error				8.9 %		26.1 %

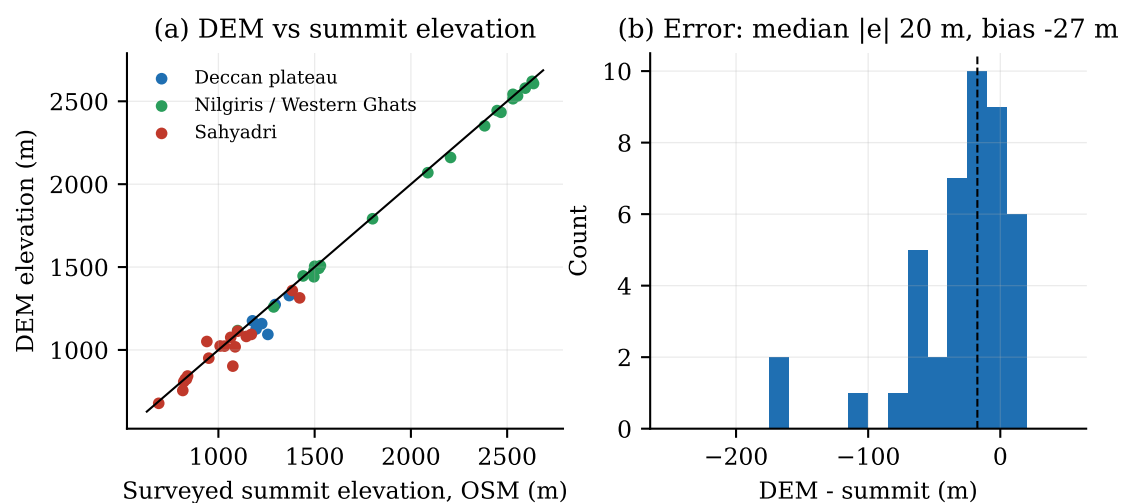
^a WMO 1991–2020 normal published by the Tokyo Climate Center from IMD observations; other stations: IMD Climatological Tables [18].

Finding that changed the design

The first version of the system used Open-Meteo with a NASA POWER fallback. NASA POWER's ~50 km cells mix the wet windward Ghats with the dry plateau behind them, so Pune reads almost twice its true rainfall. That dry plateau is exactly where pond planning matters. For Hiware Bazar, NASA POWER gave 910 mm against CHIRPS's 713 mm, and the recommended pond changed from 41,298 to 36,856 m³. CHIRPS (0.05°, calibrated against rain gauges) was therefore made the default, and a user-supplied normal can correct any remaining bias. CHIRPS still underestimates the desert (Jaisalmer, −28 %), a known weakness of satellite rainfall in arid regions. Open-Meteo could not be scored because its daily request quota for the test machine's IP had been exhausted.

7.3 Elevation

Terrain-tile elevations were compared with 44 OpenStreetMap summits that have an `ele` tag, in three regions (Figure 11). The median absolute error is 20 m, with a systematic bias of −27 m: a 30 m DEM averages a sharp summit with its flanks. Summits are therefore a worst case. Pond sites lie on gentle valley floors and depend on *relative* elevation, for which the SRTM specification is 10 m (90 %) and actual errors on gentle terrain are smaller [15]. The practical limitation is resolution: features narrower than about 30 m (bunds, drains, road embankments) are not resolved.

**Figure 11:** DEM compared with surveyed summit elevations: (a) scatter; (b) error distribution.

7.4 Land-cover classifier

The classifier was tested in four landscapes:

- semi-arid Maharashtra;
- irrigated Punjab;
- wet Kerala (Kuttanad);
- arid Rajasthan.

In each, 25 of the classifier's 20 m cells were drawn at random (seed 42). Each cell was rendered as a 100 m chip of zoom-18 imagery, about four times finer than the classifier input, with the cell outlined. The classifier's output was saved separately and not shown on the chips. Each cell was labelled by eye with its dominant cover. Cells split roughly evenly between two covers were marked mixed and excluded (13 of 100). Accuracy was assessed following Congalton [16], with Cohen's κ [17] for agreement beyond chance.

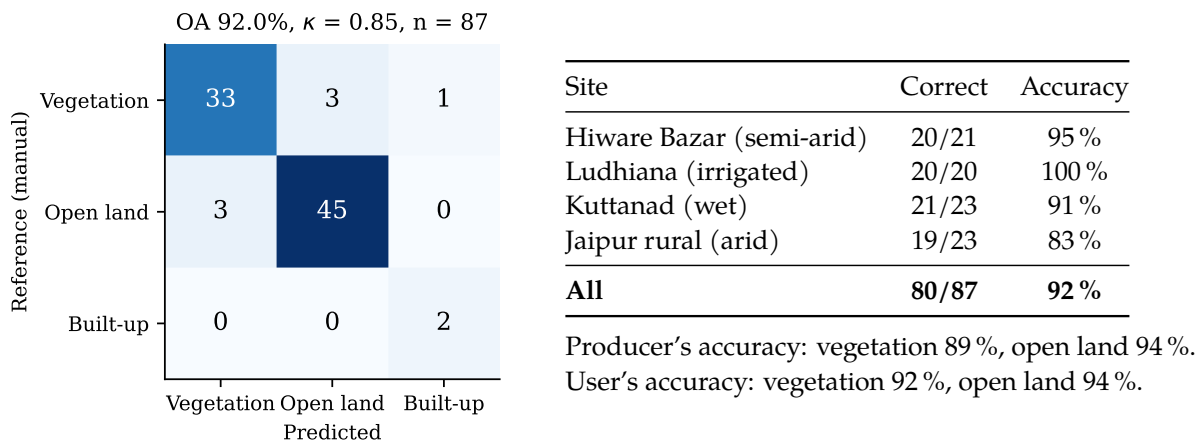


Figure 12: Land-cover confusion matrix and per-site accuracy (87 unmixed cells).

The errors fall into three groups:

- Three are pale-green sparse grass in arid Rajasthan, which GRVI calls vegetation. That is a borderline case with little consequence for siting.
- Two are cells with a single tree covering about half of the cell.
- Random sampling drew only two built-up cells and no water cells, so the accuracy of those classes is not established by this sample.

In the system, buildings, roads and water bodies are primarily excluded from OpenStreetMap with safety buffers; the satellite classes provide a secondary safeguard.

7.5 Sensitivity analysis

The sensitivity of runoff to its two most uncertain inputs was explored on the real Hiware Bazar series (Figure 13). Runoff responds to rainfall with an elasticity of about three: $\pm 10\%$ rainfall changes runoff by -30% and $+31\%$. A rational-method model would predict a proportional $\pm 10\%$. Soil group is even more influential: from group A to group D, runoff rises from 37,301 to 176,527 m^3/yr , a factor of 4.7. The two inputs that matter most are therefore the rainfall dataset, which is why it was validated and made correctable, and the soil group. The soil group is exposed

as a parameter with plain descriptions (e.g. “black cotton soil” for D) because it cannot be inferred from imagery.

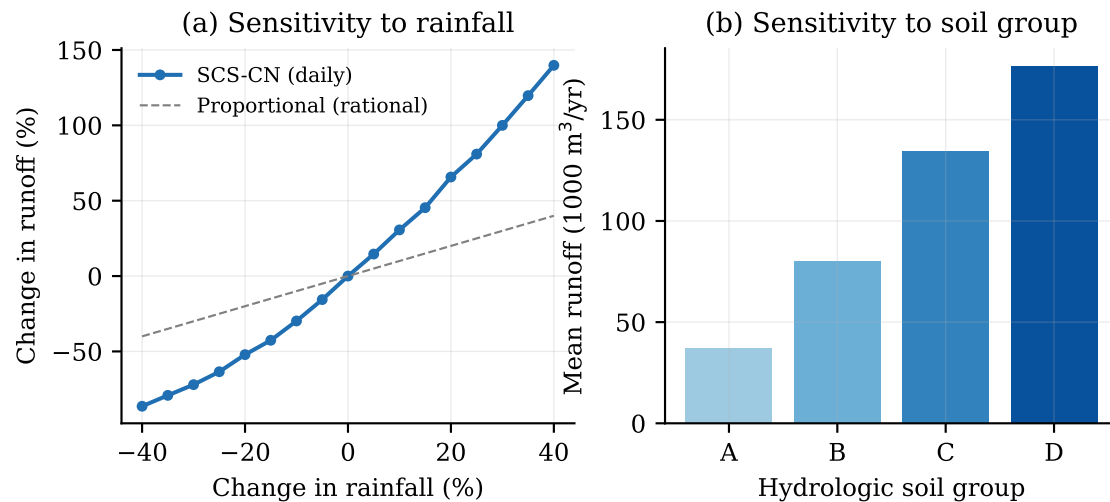


Figure 13: Sensitivity of mean annual runoff at Hiware Bazar (rank 1): (a) to a uniform scaling of daily rainfall; (b) to hydrologic soil group.

7.6 What has not been validated

- Runoff volumes against measured stream flow: no gauged village-scale catchments exist in open data.
- Catchment boundaries against official micro-watershed maps (e.g. Bhuvan).
- Legal land ownership, which is absent from open data.
- Siting against locations chosen by engineers, which would be possible with MGNREGA or Amrit Sarovar site records.

8 Software quality and project management

Modularity

Each analysis stage is a separate module with a narrow, array-based interface; the pipeline composes them. Adding a terrain source, rainfall provider or land-evidence layer touches one module.

Typing and validation

Type hints throughout; Pydantic models validate every request (ranges, enumerations) and generate the OpenAPI reference at `/docs`.

Testing

Unit tests cover all formulas and the hydrology core; validation scripts cover the data sources. Both can be re-run with one command.

Traceability

Every response carries its data sources, methods, parameters, candidate ranking rule and per-stage timings in `method`, plus any degradations in `warnings`.

Reproducibility

Pinned dependencies (`requirements.txt`, `requirements-dev.txt`), a Dockerfile, determinis-

tic random seeds in validation, and figures regenerated from saved results by `report/figures/-make_figures.py`.

Configuration

Environment variables only (MAPTILER_KEY, DATABASE_PATH, TILE_CACHE_DIR); no secrets in the code; `.env` excluded from version control.

Deployment

Container-ready (Docker) and deployable on free platforms such as Render with a persistent disk for the database.

9 Limitations and future work

- **Resolution.** The ~30 m DEM cannot see small bunds or drains. Integration with CartoDEM, drone surveys or LiDAR would sharpen siting.
- **Ownership.** Land availability is inferred. The parcel interface is the hook for integrating state revenue maps (Bhu-Naksha) or panchayat records.
- **Soils and groundwater.** Soil group is a user input. The ISRIC SoilGrids API could supply texture automatically, and groundwater depth would bound excavation depth.
- **Imagery.** RGB-only classification limits built-up and water detection. Sentinel-2 NDVI/NDWI/NDBI would improve both.
- **Rainfall.** IMD 0.25° gridded data could be ingested as an additional source; climate-change scenarios could stress-test designs.
- **Economics.** Cost per cubic metre of storage and benefit indicators (irrigable area, recharge) would support budget decisions between candidates.
- **Learning from practice.** Records of existing successful ponds could be used to learn the site-score weights instead of fixing them.

10 Conclusion

The Village Pond Planning System meets all eight functional requirements of the brief with a complete, documented, tested and validated web application. It turns a village name into a ranked shortlist of hydrologically independent pond sites. Each site comes with its catchment, twenty-year rainfall statistics, daily-simulated runoff and engineering dimensions, all on an accessible interactive map. The validation work was as important as the implementation: it exposed a 92 % rainfall error in a widely used dataset and changed the system’s design. It also quantified the accuracy of every other component, and showed that rainfall and soil type dominate the uncertainty in pond sizing. The result is a pre-feasibility tool whose outputs are transparent about their sources, methods and limits, ready to be checked in the field.

References

- [1] Ministry of Rural Development, Government of India. *Mission Amrit Sarovar: Guidelines*. 2022.
- [2] Mapzen / Amazon Web Services. *Terrain Tiles on AWS (Joerd)*. Open Data on AWS, <https://registry.opendata.aws/terrain-tiles/>.
- [3] R. Barnes, C. Lehman, D. Mulla. Priority-flood: An optimal depression-filling and watershed-labeling algorithm for digital elevation models. *Computers & Geosciences*, 62:117–127, 2014.

- [4] J. F. O’Callaghan, D. M. Mark. The extraction of drainage networks from digital elevation data. *Computer Vision, Graphics, and Image Processing*, 28(3):323–344, 1984.
- [5] C. J. Tucker. Red and photographic infrared linear combinations for monitoring vegetation. *Remote Sensing of Environment*, 8(2):127–150, 1979.
- [6] T. Motohka, K. N. Nasahara, H. Oguma, S. Tsuchida. Applicability of green-red vegetation index for remote sensing of vegetation phenology. *Remote Sensing*, 2(10):2369–2387, 2010.
- [7] N. Otsu. A threshold selection method from gray-level histograms. *IEEE Transactions on Systems, Man, and Cybernetics*, 9(1):62–66, 1979.
- [8] C. Funk et al. The climate hazards infrared precipitation with stations — a new environmental record for monitoring extremes. *Scientific Data*, 2:150066, 2015.
- [9] H. Hersbach et al. The ERA5 global reanalysis. *Quarterly Journal of the Royal Meteorological Society*, 146(730):1999–2049, 2020.
- [10] R. Gelaro et al. The Modern-Era Retrospective Analysis for Research and Applications, Version 2 (MERRA-2). *Journal of Climate*, 30(14):5419–5454, 2017.
- [11] G. H. Hargreaves, Z. A. Samani. Reference crop evapotranspiration from temperature. *Applied Engineering in Agriculture*, 1(2):96–99, 1985.
- [12] R. G. Allen, L. S. Pereira, D. Raes, M. Smith. *Crop Evapotranspiration: Guidelines for Computing Crop Water Requirements*. FAO Irrigation and Drainage Paper 56, Rome, 1998.
- [13] USDA Soil Conservation Service. *Urban Hydrology for Small Watersheds*. Technical Release 55, 2nd ed., 1986.
- [14] R. H. Hawkins, A. T. Hjelmfelt, A. W. Zevenbergen. Runoff probability, storm depth, and curve numbers. *Journal of Irrigation and Drainage Engineering*, 111(4):330–340, 1985.
- [15] T. G. Farr et al. The Shuttle Radar Topography Mission. *Reviews of Geophysics*, 45(2):RG2004, 2007.
- [16] R. G. Congalton. A review of assessing the accuracy of classifications of remotely sensed data. *Remote Sensing of Environment*, 37(1):35–46, 1991.
- [17] J. Cohen. A coefficient of agreement for nominal scales. *Educational and Psychological Measurement*, 20(1):37–46, 1960.
- [18] India Meteorological Department. *Climatological Tables of Observatories in India 1991–2020*. IMD, Pune.

A Installation guide

Requirements. Python 3.12 or newer and an internet connection. No API keys are required.

```
# 1. create an isolated environment and install dependencies
python3 -m venv .venv
.venv/bin/pip install -r requirements.txt # add -dev for pytest

# 2. (optional) configuration
cp .env.example .env # MAPTILER_KEY, DATABASE_PATH, TILE_CACHE_DIR

# 3. run
.venv/bin/uvicorn app.main:app --port 8000
# app: http://127.0.0.1:8000 API docs: http://127.0.0.1:8000/docs

# 4. tests and validation
.venv/bin/python -m pytest -q tests
.venv/bin/python validation/rainfall_check.py # etc.
```

```
# Docker
docker build -t pond-planner .
docker run -p 8000:8000 -v pond-data:/app/data pond-planner
```

B API reference

The interactive OpenAPI documentation is served at /docs. Table 10 summarises the routes.

Table 10: HTTP API.

Method	Route	Purpose
POST	/api/analyze	Full analysis for a location (JSON body; only lat, lon required)
POST	/analyzeContour	Full analysis from an uploaded KML/KMZ (multipart; alias /findCatchment)
GET	/api/geocode?q=	Village search (Nominatim)
GET	/api/reverse?lat=&lon=	Place name for a point
GET	/api/rainfall?lat=&lon=	Rainfall statistics only
	&years=&source=	
GET	/api/analyses	Saved analyses (summaries)
GET/DELETE	/api/analyses/{id}	Retrieve or delete one analysis
GET	/health	Liveness probe

Request parameters of /api/analyze.

Field	Default	Meaning
lat, lon	—	Centre of the analysis area
village	null	Label stored with the analysis
radius_m	2000	Half-width of the square area (500–6000)
grid_resolution	200	Cells per side (60–320)
n_candidates	5	Ranked sites to return (1–10)
pond_lat, pond_lon	null	Optional user-chosen site (rank 1, snapped)
land_parcel	null	GeoJSON of government land; siting restricted to it
soil_group	C	NRCS hydrologic soil group A–D
rainfall_source	auto	auto, chirps, open_meteo, nasa_power
annual_normal_mm	null	Known local normal; the series is scaled to it
rainfall_years	20	Length of record (5–40)
fill_fraction	0.5	Storage as a fraction of dependable runoff
side_slope, max_depth_m	1.5, 4.5	Excavation geometry limits
max_slope_deg	8	Steepest ground allowed for excavation

Response. The response contains:

- pond_site, catchment, runoff, pond_design and storage for the top site;
- candidates, which holds the same objects for every ranked site;
- rainfall statistics and the land assessment;
- layers, the GeoJSON and PNG overlays;
- warnings, method, processing_seconds and analysis_id.

Errors have the form {"status": "error", "detail": "..."} with status 400, 413, 422 or 502.